complaint-router

Routing consumer financial complaints to the right team, with a system that keeps learning

KSE · Introduction to MLOps and ML System Design · Lab 1 · Zakhar Staryk · October 2026

github.com/goodfrog33/kse-mlops-complaint-router · tag lab-1

18.5 M
complaints in the CFPB database

3.8 M
of them with complaint text

10
product queues to route to

At a glance

What the project is, which data it uses and whether it fits the five labs, on one page.

The problem

Consumers file about 21,000 complaints a day about financial products with the US regulator CFPB. Each must reach the team that owns the product; a misrouted complaint waits days while the 15-day response clock runs.

What the system does

Reads the complaint text and predicts one of 10 product queues with a calibrated confidence. Confident → routed automatically; otherwise → a human, with the prediction as a suggestion. Retrained as the data drifts, monitored before labels arrive.

The data

CFPB Consumer Complaint Database (public domain). Official daily export: 18.5 M complaints, no text since 2026. July 2026 snapshot: 3.8 M complaint texts (2015–2025), replayed month by month. Profiled in docs/data-profile.md; 200 real rows in data/sample/.

Fit with the five labs

Yes. Each lab has real material: data pipeline and MLflow (Lab 2), LLM baseline and retrieval (Lab 3), serving and real drift (Lab 4), an operations agent (Lab 5). One caveat: no new complaint text after 2025 (slide 22).

How to read: 3–6 the problem and requirements · 7–15 the data, with real examples · 16–21 the design · 22–23 the fit with the labs · 24 where everything is in the repository.

Every complaint has to reach the team that owns it

A misrouted complaint waits days; confident cases go straight to the right team, the rest to a human.



Service down or > 2 s → human triage. Intake never stops.

Who consumes

The intake desk's ticket queue calls POST /predict once per complaint; agents see the suggestion when it is not auto-routed.

How often

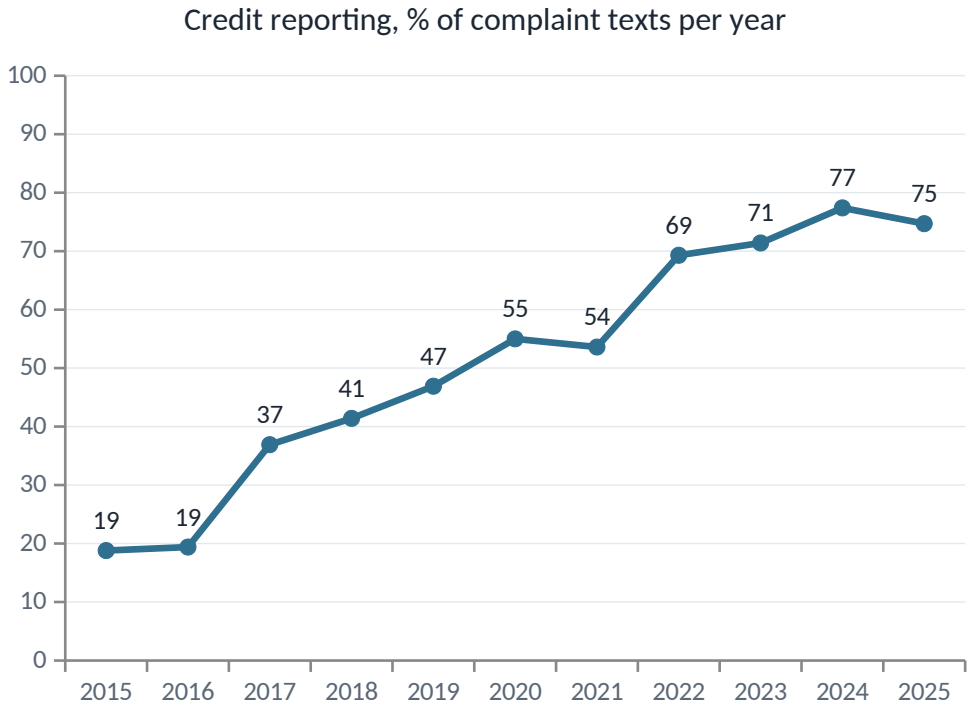
~21,000 complaints a day in 2026 (measured in the official export); peak ≈ 2.5 requests/s.

Cost of an error

Wrong team → reassignment while the 15-day response clock runs. Systematic errors hide until labels arrive.

The data keeps moving, so a one-off notebook would decay

Labels, class mix, volume and published fields all changed (measured), so the model must be retrained and watched.



Label set changed twice

April 2017 and August 2023 renamed and merged products; 20 names now map to 10 classes.

Class mix shifted

Credit reporting: 19% of texts (2016) → 77% (2024).

Volume grew 30×

0.17 M complaints (2015) → 5.4 M (2025).

Text stopped in 2026

CFPB no longer publishes the complaint narrative.

Success is measured on three separate layers

Business, model and system metrics are kept apart; two are optimised, the rest are thresholds a new model must pass first.

Business

≤ 8%
complaints routed to the wrong queue
optimizing · assumed baseline 15%

≥ 60%
routed with no human touch
satisficing

Model

≥ 0.80
macro-F1, newest held-out month
optimizing · ≥ baseline + 0.10

≥ 0.95
precision on auto-routed complaints
satisficing · sets the threshold

System

≤ 300 ms
p95 latency of POST /predict
satisficing

≥ 99.5%
availability, 08:00–20:00
satisficing

≤ 3 days
age of the newest ingested data
satisficing

Macro-F1, not accuracy: credit reporting is 74% of training texts, so "always credit reporting" would score well on accuracy and be useless for the other nine queues.

Every non-functional requirement carries a number

One row per category of the design-doc template; where two requirements conflict, the slide says which one wins and why.

Category	Requirement
Freshness	Structured export ingested daily, ≤ 3 days old · text replayed monthly · labels joined after 15 days
Latency, throughput	p95 ≤ 300 ms, p99 ≤ 800 ms · 20 req/s per replica vs ~2.5 req/s peak
Availability, fallback	≥ 99.5% · > 2 s or error → human queue · failed load keeps previous champion
Reproducibility	Any model = git commit + DVC data version + params, rebuilt by one command
Adaptability	Monthly and alert-triggered retraining · unmapped product fails the quality gate
Observability w/o labels	Input, prediction and confidence drift for the 15 days before labels
Cost, resources	Whole stack in Minikube, 4 GB / 2 CPU · model ≤ 200 MB · cloud ≤ $800/month

Conflict 1

Freshness vs reproducibility → reproducibility wins. Train only on DVC snapshots; costs ≤ 1 day.

Conflict 2

Precision vs coverage → precision wins. A wrong auto-route costs more than one more manual review.

One raw row, exactly as CFPB publishes it

Each field has one role: model input, label, join key, or excluded because it only exists after routing (target leakage).

Field	Value (complaint 11425997)	Role
Date received	2025-01-07	time split, replay
Product	Mortgage	LABEL
Sub-product · Issue	Conventional home mortgage · Trouble during payment process	not used (5.1)
Company	Shellpoint Partners, LLC	feature · skew risk
State · Submitted via	WI · Web	feature
Consumer complaint narrative	“Property Taxes in Escrow did not get paid to my village by the due date, NewRez assures they will automatically be paid and not to worry about it. I now am in the new tax year and they are still listed as unpaid…”	MAIN FEATURE
Date sent to company	2025-01-07	excluded: leakage
Company response	Closed with explanation	excluded: leakage
Complaint ID	11425997	join key for labels

A sample of the data: one real complaint per class

Ten rows from data/sample/complaints_sample.csv (January–June 2025). The class is the label; the text is cut to its first words here.

Complaint ID	Class (label)	Company	Narrative (start)
13456120	**Checking or savings**	KeyCorp	*I am disputing a negative account reported by [KeyBank…] that is blocking me from opening any bank accounts…*
14373622	**Credit or prepaid card**	Synchrony Financial	*I submitted proof of product return as requested. I received letter/s from Synchrony Bank…*
12867215	**Credit reporting**	TransUnion	*I am a victim of XXXX. The information listed below, which appears on my credit report, is the result of that…*
13671992	**Debt collection**	Experian	*There are collection accounts on my report that I believe contain inaccurate information…*
11529549	**Debt or credit mgmt**	TransUnion	*I requested proof of late payments. I requested the late payments be investigated…*
11650859	**Money transfer**	Block, Inc.	*A XXXX company charged my card almost {$200.00}… I contacted Cash app and they refuse to give my money back…*
11501270	**Mortgage**	Shellpoint Partners	*I have paid more than XXXX XXXX of my mortgage… I suppose to stop paying PMI, but the Newrez refused…*
12279657	**Payday / personal loan**	Regional Management	*I applied for a loan @ Regional Finance. They changed the approved loan amount & terms 3 different times…*
13302787	**Student loan**	MOHELA	*Report servicer misconduct, including errors in loan balances, misapplied payments, and wrongful denials of repayment plans…*
13152318	**Vehicle loan or lease**	Ford Motor Credit	*I set this account up on autopay… for some reason it was not set…*

What the model reads: real complaints and their labels

The product is usually clear from the consumer's own words; XXXX is CFPB's redaction of personal data (in 62% of texts).

Mortgage — Shellpoint · WI · 2025-01-07

"Property Taxes in Escrow did not get paid to my village by the due date, NewRez assures they will automatically be paid and not to worry about it…"

Student loan — MOHELA · OH · 2025-01-16

"I have student loans being serviced by MOHELA. I have been on the SAVE plan. Once SAVE plan users were placed on general forbearance, interest should not have been accruing…"

Money transfer — Block, Inc. · IL · 2025-01-18

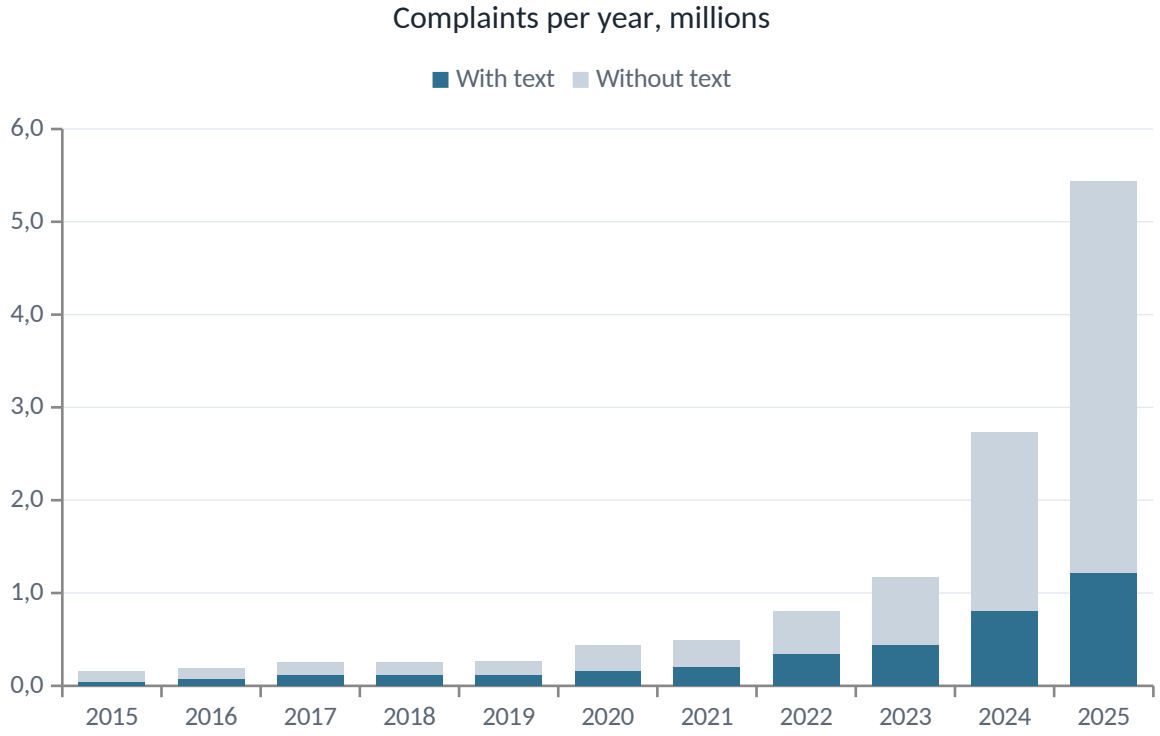
"A XXXX company charged my card almost {$200.00}. When I called to let them know they told me they were processing refund but didnt so I contacted Cash app and they refuse to give my money back…"

Credit reporting — Equifax · CT · 2025-02-19

"I am writing to formally address an urgent issue with my credit report. It has come to my attention that certain information associated with my account has been incorrectly mixed with that of another individual…"

3.8 million complaints with text, 2015 to 2025

The July 2026 snapshot holds 16.6 M complaints; 3.8 M carry the consumer's text, far more than the system needs.



16.6 M

complaints in the July 2026 snapshot

3.8 M

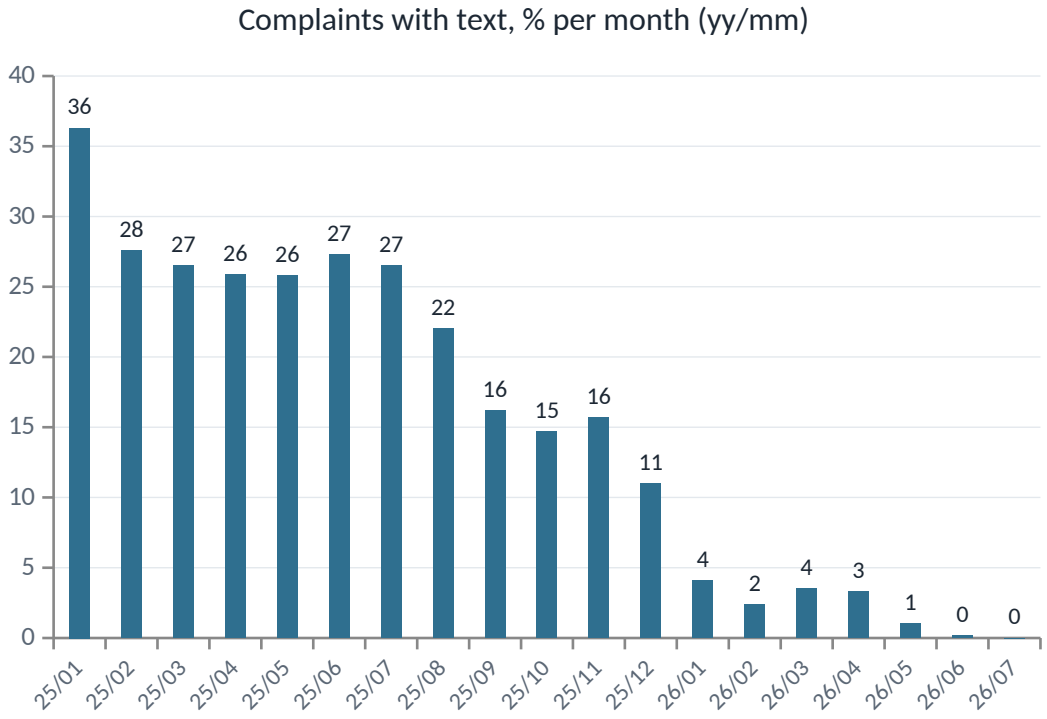
with the consumer's own text

1.5 M

texts in the 24-month training window

Finding: CFPB stopped publishing complaint text in 2026

The most important data finding: text up to 2025 is frozen, so it is versioned once and replayed month by month.



What changed

The official export (2026-10-17) has 15 columns and no narrative. 26% of complaints had text in 2025-07; 0% in 2026-07.

What the design does

- Text: 2026-07 snapshot, put in DVC once, replayed one month per run
- Daily structured export: freshness and volume checks
- Quality gate “text share < ½ reference” fires on 2025-12

The label set changed twice; a versioned map absorbs it

CFPB used three product taxonomies; a versioned table maps all 20 product names onto 10 stable classes.

Dec 2011 – Apr 2017

- Credit reporting
- Bank account or service
- Consumer Loan
- Money transfers

Apr 2017 – Aug 2023

- Credit reporting, credit repair services, or other personal consumer reports
- Credit card or prepaid card
- Checking or savings account

Aug 2023 – today

- Credit reporting or other personal consumer reports
- Payday loan, title loan, personal loan, or advance loan
- Debt or credit management (new)

→

10 classes

20 CFPB names → 10 stable classes

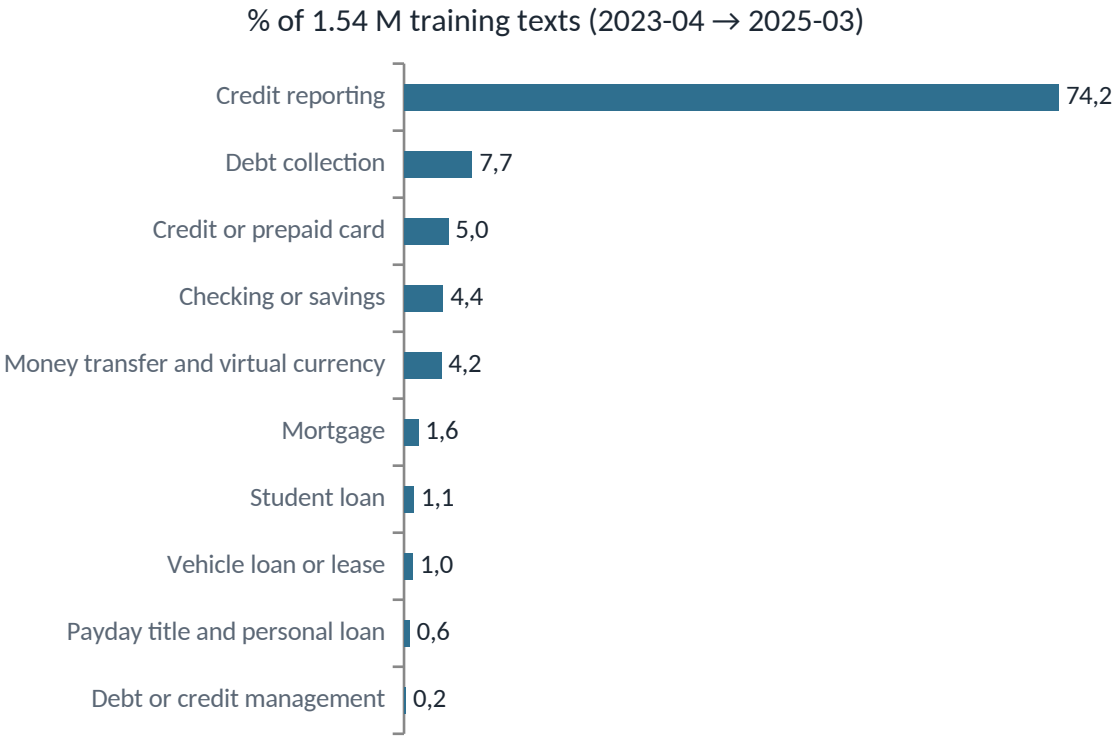
configs/ taxonomy_map.csv

unmapped → manual triage + gate fails

Every era boundary is visible in the data: product names start and stop exactly in 2017-04 and 2023-08 (docs/data-profile.md).

Training texts are heavily imbalanced

Credit reporting is three quarters of the training data, which decides the main metric and the promotion rule.



Macro-F1

every queue weighs the same in the main metric

Class weights

small queues are not ignored in training

Recall guard

no class may lose > 3 pp in a new model

Labels are noisy: two real examples from the sample

Consumers choose the product themselves and sometimes choose the wrong one; the design measures this instead of ignoring it.

Labelled: Debt collection

TransUnion · VA · 2025-04-09

“My credit report shows collection accounts that I believe include inaccurate information. Under my rights as outlined in 15 USC 1681e (b) and 15 USC 1681i, I am entitled to a fully accurate report…”

Labelled: Debt or credit management

TransUnion · WI · 2025-01-14

“I requested proof of late payments. I requested the late payments be investigated and that TransUnion send me there method of verification…”

What the design does about it

- Lab 2: hand-label 200 rows and report the noise rate next to the baseline
- Human overrides take precedence over the CFPB label; the model's own predictions are never used as labels
- Macro-F1 targets are read as “against noisy labels”: 0.80 is a floor, not a ceiling

The data passes all four checks the labs require

The assignment's data checklist, the evidence for each item, and the time split the data allows.

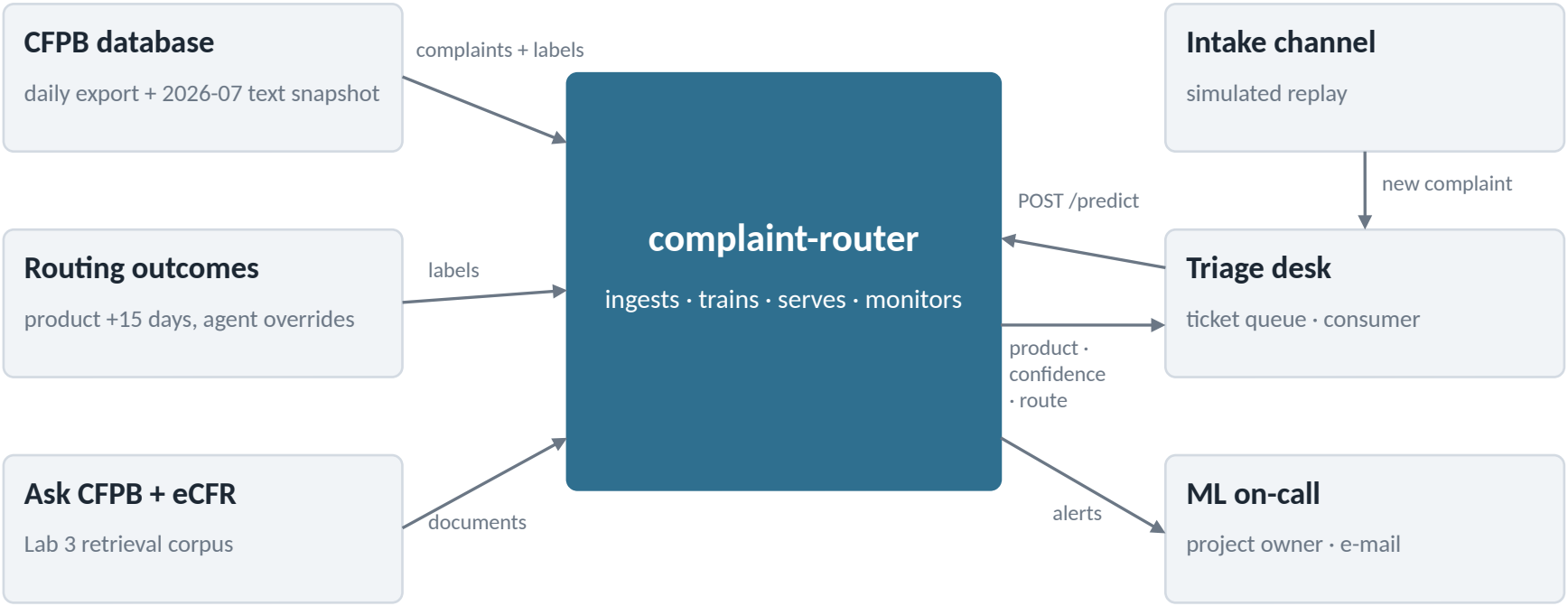
PASS	**Time axis**	Daily date_received, 2011-12 → today; monthly batches replayed from the snapshot
PASS	**History to hold out**	3.8 M texts; train 2023-04 → 2025-03, validate 2025-04 → 06, test 2025-07 → 09
PASS	**Label arrival delay**	15 days (emulated), grounded in CFPB's rule: published after the company responds or after 15 days
PASS	**Text surface / corpus**	Narratives are the data; Lab 3 corpus: Ask CFPB articles + eCFR Regs B, E, F, V, X, Z



Split by time, never random: complaints about one event share wording and cluster in time.

System context: one box, every external party around it

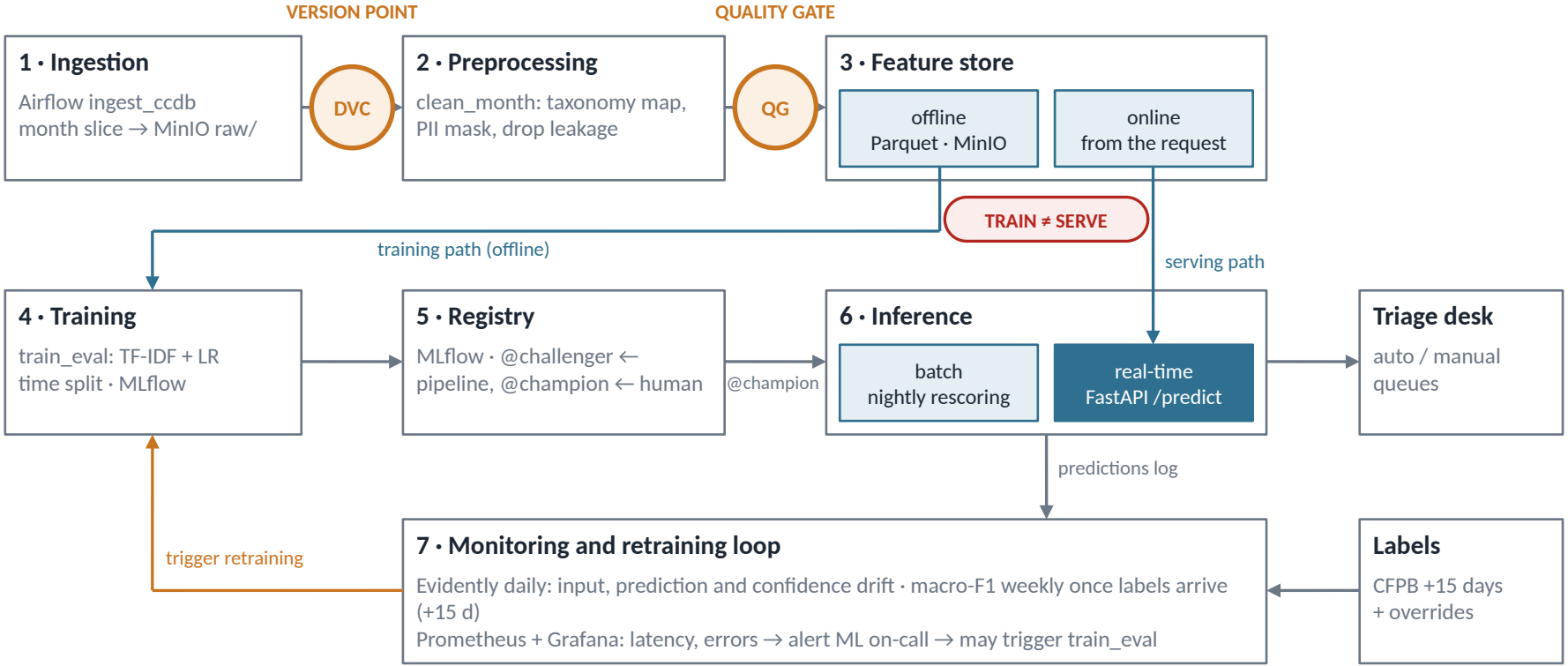
Who sends data, who consumes predictions, who gets alerts; names match every diagram and design.md §6.2.



One box. Everything outside it belongs to someone else.

Data flow on the seven layers, with the three marks

Orange circles: data version (DVC) and quality gate. Red tag: where training and serving paths diverge.



Model: start from rules, keep the candidate simple

Two baselines, one simple candidate, and for every feature the place its value comes from at serving time.

Baseline 0

always “Credit reporting”: the macro-F1 floor

Baseline 1

keyword rule per product; no match → manual

Candidate

TF-IDF (word 1–2, char 3–5) + one-hot fields → logistic regression, class weights, isotonic calibration

Feature	Training source	Serving source
Narrative TF-IDF	clean/ Parquet	request text, same preprocess()
Narrative length	clean/ Parquet	computed from request
Submitted via, state	CFPB fields	request; missing → unknown
Company bucket	CFPB normalized name	raw text → alias table (skew risk, monitored)

LLM zero-shot classifier: evaluated as the LLM baseline in Lab 3, not as the router.

CFPB masks personal data as XXXX in 62% of texts; the service applies the same mask at intake.

Monitoring: what we can see before the labels arrive

Labels arrive 15 days after a prediction, so the first two weeks rely on proxy signals; quality is measured once labels land.



Visible from day 0 (proxies)

- Prediction drift: PSI of class mix > 0.2
- Input drift: OOV-token rate +50%, text length, channel mix
- Confidence: mean max-prob −0.05, auto share < 50%
- Skew: unmatched company names > 15%
- Service: p95 > 300 ms or 5xx > 1% for 5 min
- Data: newest row > 3 days old; row count equal to previous interval; text share < ½ reference

From day 15 (labels)

- macro-F1 < 0.75 or a class recall −10 pp → alert + retrain trigger
- Real misrouting rate (business metric)
- Concept drift: same words, different correct product

Alerts: Grafana e-mail to the ML on-call, acknowledged within 1 business day.

Runs on a laptop; every component has a priced cloud twin

Everything runs locally (Docker Compose, then Minikube); the AWS column is named and priced as the template requires, not run.

Local (pinned)	AWS equivalent	$ / month
MinIO (S3 API)	Amazon S3	3
PostgreSQL 16	Amazon RDS	15
Airflow 3.3.1	Amazon MWAA	250
MLflow 3.16.1	SageMaker with MLflow	150
FastAPI on Minikube	Amazon EKS + ECR	136
Prometheus + Grafana	CloudWatch	15
Evidently	EKS CronJob	0
pgvector (Lab 3)	OpenSearch Serverless	175
LLM (Lab 3)	Amazon Bedrock	160

$904

estimated AWS cost per month

$800

ceiling set in section 4

$729

with pgvector in RDS instead of OpenSearch

Nothing is run in the cloud; naming and pricing is the requirement.

Known risks, each with its answer

Each risk has a mitigation and the lab that implements it; the first one has already happened.

Text source stopped (2026)

Versioned snapshot in DVC; monthly replay

Lab 2

Taxonomy change

taxonomy_map.csv; unmapped product fails the gate

Lab 2

Training/serving skew

preprocess() inside the model; unmatched-name alert

Labs 2, 4

15 days without labels

Drift, confidence and override proxies

Lab 4

4 GB laptop cluster

Linear model, LocalExecutor, hosted LLM

Lab 4

Feedback loop

Labels from CFPB and overrides, never from predictions

Lab 2

This project carries all five labs

For each lab: what it needs, what this project provides, and the verdict.

Lab	What the lab needs	What this project provides	Fit
1 · Design	Use case, data justification, 3 diagrams, repo with CI	design.md, data profile, 200-row sample, taxonomy map, green CI, tag lab-1	**Done**
2 · Data + MLflow	Batches over time, held-out period, a bad batch for the gate	Monthly replay of 3.8 M texts; time split; a real bad batch: 2026 months without text	**Fits**
3 · LLM + retrieval	A task for an LLM baseline; a corpus to retrieve from	Zero-shot LLM vs TF-IDF on 1,000 test texts; Ask CFPB + eCFR for cited rationales	**Fits**
4 · Serving + drift	Servable model, real drift, known label delay	Small model on FastAPI; replay of 2023-08 taxonomy change and the credit-reporting surge; 15-day delay	**Fits**
5 · Agent	Operational tasks an agent can do and be scored on	Read drift reports, compare challenger vs champion (5.4 rule), draft the promotion note	**Fits**

One caveat, stated in design.md §5.2 and §6.9: no new complaint text after 2025, so “new” text batches are replayed history. The daily structured export still exercises freshness and volume checks.

How the next four labs grow this system

Each lab revises numbered sections of design.md and adds rows to the README maturity scorecard, so nothing starts over.

Lab 2

Data pipeline, versioning, MLflow

Airflow DAG ccdb_pipeline: ingest → DVC snapshot → clean → quality gate → train. Baselines and TF-IDF model tracked in MLflow; @challenger registered.

Proves

Data versioned · pipeline idempotent · quality gate (a 2026 month without text fails it) · experiments tracked · model with lineage

Revises §5.2 5.4 6.1 7.2

Lab 3

LLM baseline and retrieval

Zero-shot LLM classifier scored against TF-IDF on 1,000 test texts. pgvector index of Ask CFPB + eCFR; cited rationales for manual-queue complaints.

Proves

LLM component evaluated · prompts versioned · tokens, cost, p50/p95 recorded

Revises §5.3 6.2 6.7

Lab 4

Serving, monitoring, drift

API and model images on Minikube with health probes; nightly Evidently; Grafana. Drift replay: the 2023-08 taxonomy change and the credit-reporting surge.

Proves

Services containerized · health contracts · monitoring scheduled · drift alert reaches a human · one-command start

Revises §6.1 6.3 6.6 7.3

Lab 5

Agentic extension

An agent reads drift reports, compares @challenger with @champion by the §5.4 rule and drafts the promotion note; a human approves.

Proves

Agent traced and evaluated: ten tasks, run twice

Revises §6.8 7.4

Where to find everything

Everything here can be checked and reproduced from github.com/goodfrog33/kse-mlops-complaint-router (tag lab-1).

Path	What it is
docs/design.md	Design document, sections 1–7.4, with the three diagrams
docs/data-profile.md	All data numbers in this deck, by year, month, product and class
scripts/profile_data.py	Recomputes the profile and the sample from the raw files
data/sample/complaints_sample.csv	200 real complaints, 20 per class
configs/taxonomy_map.csv	20 CFPB product names → 10 classes
pyproject.toml, uv.lock, Makefile	Pinned environment; make setup, make lint
.pre-commit-config.yaml, .github/workflows/	The same linter on every commit and push

Check it

```
git clone …
make setup
make lint
```

Both pass on a clean clone; CI runs make lint on every push.

Reproduce the data numbers

Download steps in docs/data-profile.md, then
uv run python scripts/profile_data.py